MINI PROJECT REPORT

24CS503 - Machine Learning

Project Title:
Hospital Patient Segmentation using K-Means and DBSCAN

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CERTIFICATE

This is to certify that the mini project report entitled "Hospital Patient Segmentation using K-Means and DBSCAN" is a bonafide record of the work carried out by Tamil Mani B, A Vishnu Vardhan and Yuvaraj U of CSE - B under my guidance and supervision as part of the requirements of the Machine Learning course during the academic year [2026–2027].

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DECLARATION

We hereby declare that the mini project entitled "Hospital Patient Segmentation using K-Means and DBSCAN" is our original work carried out as part of the Machine Learning course. The dataset sources, algorithms, tools, references, and contributions of other authors have been appropriately acknowledged. The work has not been submitted elsewhere for any other academic award.

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ABSTRACT

Hospitals treat patients whose clinical profiles differ widely, and grouping similar patients can support screening and follow-up planning. This project applies unsupervised machine learning to segment patients using the Pima Indians Diabetes Database, which has 768 patient records and eight clinical attributes. The diabetes label was not used to form clusters; it was used only afterwards to validate them. Impossible zero values in glucose, blood pressure, skin thickness, insulin and BMI were treated as missing and replaced by the median, and all features were standardised. K-Means (k = 2 to 8) and DBSCAN (a grid of eps and min_samples values) were compared using inertia, the silhouette score, the Davies-Bouldin index, the Calinski-Harabasz index and the adjusted Rand index. K-Means with k = 3 gave three stable groups of 248, 335 and 185 patients (mean adjusted Rand index 0.991 over ten random starts) with diabetes rates of 52%, 13% and 52%, against 34.9% overall. The silhouette score was only 0.195, which shows that the groups overlap and are not sharply separated. DBSCAN found one dense group and 88 outlying patients, so it behaved as an outlier detector and not as a segmentation method. A Streamlit web application was built to explore the clusters and to place a new patient in the most similar group. K-Means was the more useful method for patient segmentation on this data. The groups describe similar measurements and are not diagnoses.

Keywords: Patient segmentation, K-Means, DBSCAN, Clustering, Pima Indians Diabetes

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1. INTRODUCTION

1.1 Background

Healthcare organisations record clinical measurements such as blood glucose, blood pressure, body mass index and age for every patient. Patients with similar measurements often need similar care, but grouping them by hand is slow when there are many patients and many attributes. Machine learning, in particular unsupervised clustering, can group patients automatically from the data without any predefined labels. Diabetes is a useful test case because it is a common long-term condition in which several measurements influence risk together.

1.2 Motivation

Hospitals have limited time and resources. If groups of patients with different risk profiles can be found, screening, counselling and follow-up can be planned for each group. Doctors, hospital administrators and public-health planners may benefit. A second motivation is to learn how clustering behaves on real, imperfect medical data that has missing values and overlapping groups.

1.3 Scope of the Project

Included: unsupervised segmentation of 768 patients using eight clinical attributes; data cleaning and scaling; K-Means and DBSCAN; evaluation with internal measures; validation of the groups against the diabetes label after clustering; and a web application to explore the results. Excluded: diagnosing or predicting diabetes, supervised models, clinical validation by doctors, other datasets, and deployment in a hospital.

1.4 Applications

Practical applications of patient segmentation include:

- Targeted screening and awareness programmes for each patient group.
- Planning of follow-up visits and hospital resources by group size and risk.
- Spotting unusual patients whose measurements differ from most others.

- A starting point for later supervised models and feature studies.

2. PROBLEM DEFINITION AND OBJECTIVES

2.1 Problem Statement

To divide the 768 patients of the Pima Indians Diabetes Database into groups with similar clinical profiles using K-Means and DBSCAN, without using the diabetes label, and to measure how well separated and how stable the groups are and whether they differ in diabetes rate.

2.2 Objectives

1. To clean and prepare the clinical data (missing values and scaling) so that distance-based clustering is meaningful.
2. To apply K-Means for k = 2 to 8 and DBSCAN over a grid of eps and min_samples values.
3. To evaluate the clusters using the silhouette score, Davies-Bouldin index, Calinski-Harabasz index and stability across random starts, and to compare the two methods.
4. To describe each group, check it against the diabetes label after clustering, and present the results in a web application.

2.3 Expected Outcome

A small number of patient groups with clearly described average profiles, quality scores for each method, a comparison of K-Means and DBSCAN, an honest statement of how well the groups are separated, and a working web application.

3. LITERATURE SURVEY

The sources below describe the dataset, the two clustering methods and the evaluation measures used in this project.

S. No.	Author / Year	Title	Method Used	Key Finding / Limitation
1	Smith et al., 1988	Using the ADAP learning algorithm to forecast the onset of diabetes mellitus	ADAP, an adaptive learning algorithm	Introduced the Pima dataset. Predicts diabetes onset (supervised) and does not group patients. Limited to one population.
2	MacQueen, 1967; Lloyd, 1982	Some methods for classification and analysis of multivariate observations; Least squares quantization in PCM	K-Means	Simple and fast. Needs k in advance, assumes roughly round clusters, and is sensitive to scaling and outliers.
3	Ester et al., 1996	A density-based algorithm for discovering clusters in large spatial databases with noise	DBSCAN	Finds clusters of any shape and marks noise. One global eps struggles when densities differ, and results depend on eps and min_samples.
4	Rousseeuw, 1987	Silhouettes: a graphical aid to the interpretation and validation of cluster analysis	Silhouette coefficient	Label-free measure of cohesion and separation from -1 to 1. Values near 0 mean overlapping clusters.
5	Davies and Bouldin, 1979	A cluster separation measure	Davies-Bouldin index	Lower values mean more compact and better separated clusters. Assumes roughly spherical clusters.

4. DATASET DESCRIPTION

4.1 Dataset Source

Platform: Kaggle

Dataset Name: Pima Indians Diabetes Database (originally from the National Institute of Diabetes and Digestive and Kidney Diseases)
Dataset URL: https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database
Access Date: 31-10-2026

4.2 Dataset Characteristics

Characteristic	Details
Number of Instances	768 patients (women aged 21 or older, of Pima Indian heritage)
Number of Attributes	9 (8 input features and 1 outcome column)
Target Variable	Outcome (1 = diabetes, 0 = no diabetes). Not used for clustering; used only afterwards to validate the groups.
Data Type	Numerical (integer and decimal)
Missing Values	No blank cells, but zeros that cannot be real values: Glucose 5, BloodPressure 35, SkinThickness 227, Insulin 374, BMI 11. These were treated as missing. Duplicate records: 0.
Class Distribution	500 patients without diabetes (65.1%) and 268 with diabetes (34.9%)

4.3 Data Dictionary

Feature Name	Data Type	Description	Role
Pregnancies	Integer	Number of times pregnant	Input
Glucose	Integer	Plasma glucose concentration 2 hours after an oral glucose tolerance test	Input
BloodPressure	Integer	Diastolic blood pressure (mm Hg)	Input
SkinThickness	Integer	Triceps skin fold thickness (mm)	Input
Insulin	Integer	2-hour serum insulin (mu U/ml)	Input
BMI	Decimal	Body mass index (weight in kg / height in m squared)	Input
DiabetesPedigreeFunction	Decimal	Score of diabetes in the family history	Input
Age	Integer	Age in years	Input
Outcome	Integer (0/1)	Diabetes diagnosis	Target (validation only)

Note: the file used (768 rows, 9 columns) is a copy of the standard version of this dataset. Please compare it with the Kaggle download before submission.

5. METHODOLOGY

5.1 Overall Workflow

Problem Definition → Data Collection → Data Understanding → Data Preprocessing → Exploratory Data Analysis → Feature Engineering → Model Training → Model Evaluation → Interpretation → Conclusion

5.2 Data Preprocessing

- Handling missing values: zeros in Glucose, BloodPressure, SkinThickness, Insulin and BMI were treated as missing and replaced with the column median.
- Removing duplicates: checked, and no duplicate records were found.
- Outlier treatment: no records were removed. Extreme values (for example insulin up to 846) were kept, and DBSCAN was used to flag unusual patients.
- Encoding categorical variables: not required, because all attributes are numerical.
- Feature scaling / normalization: standardisation (mean 0, standard deviation 1) with StandardScaler, because K-Means and DBSCAN use distances.
- Train-validation-test split: not applicable. Clustering is unsupervised, so all 768 patients were used. The Outcome label was used only after clustering.

5.3 Algorithm Description

K-Means divides the patients into k groups by repeatedly assigning each patient to the nearest cluster centre and moving each centre to the mean of its group. We used k-means++ initialisation, 10 random starts for each k and random_state = 42, and tried k = 2 to 8. Stability was tested by repeating K-Means with ten different random seeds and comparing the results with the adjusted Rand index.

DBSCAN groups patients who lie in dense regions and marks isolated patients as noise. It has two settings: eps (neighbourhood radius) and min_samples (patients needed to form a dense region). We searched eps from 1.0 to 3.0 in steps of 0.1 with min_samples of 5, 8, 12 and 16, and used a k-distance plot with min_samples = 16 to understand the scale of eps. A setting was accepted only if it gave at least two clusters, under 25% noise and every cluster holding at least 5% of the patients. No setting passed, so the final run used eps = 2.0 and min_samples = 16.

Principal component analysis (PCA) was used only to draw the clusters in two dimensions. The clustering itself used all eight standardised features.

5.4 Evaluation Metrics

All measures except the last do not need the diabetes label, which suits unsupervised learning.

Metric	Formula / Definition	Reason for Selection
Inertia	Sum of squared distances from each patient to its cluster centre	Used for the elbow plot. It always falls as k rises, so it cannot choose k alone.
Silhouette Score	s = (b - a) / max(a, b), where a is the	Label-free measure of how well

	mean distance to the patient's own cluster and b the mean distance to the nearest other cluster. Range -1 to 1.	separated the clusters are.
Davies-Bouldin Index	Average over clusters of the largest ratio of (spread of two clusters) to (distance between their centres). Lower is better.	Compares compactness with separation.
Calinski-Harabasz Index	Ratio of between-cluster dispersion to within-cluster dispersion. Higher is better.	Second check on separation.
Adjusted Rand Index (ARI)	Agreement between two clusterings, corrected for chance. 1 = identical, 0 = random.	Used for stability across random starts and for comparing K-Means with DBSCAN.
Diabetes rate per cluster	Share of patients with Outcome = 1 in each cluster	External check, done after clustering, of whether groups carry clinical meaning.

6. IMPLEMENTATION

6.1 Software and Hardware Requirements

Category	Tool / Specification	Purpose
Programming Language	Python 3	Implementation
Development Environment	Python scripts run from the PowerShell terminal; web browser (Google Chrome) for the web application	Coding, running and demonstration
Libraries	NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, Plotly, Streamlit, python-docx	Data processing, modelling, charts and web application
Hardware	Standard Windows laptop or PC. No GPU needed because the dataset has only 768 rows.	Execution environment

6.2 Program Modules

Module	Description	Input	Output
Data loading and cleaning	Reads the CSV, treats impossible zeros as missing, fills them with the median	diabetes.csv	Clean feature table
Preprocessing	Standardises all eight features	Clean feature table	Standardised matrix
K-Means module	Runs k = 2 to 8, computes metrics, fits k = 3, tests stability	Standardised matrix	Labels, metric table, profiles
DBSCAN module	Draws k-distance plot, searches eps and min_samples, fits the final model	Standardised matrix	Labels, noise count, grid table
Visualisation and profiling	PCA plots, profile heatmap, diabetes rate per cluster	Labels and features	Figures and CSV results
Web application (Streamlit)	Five pages: dataset, K-Means, DBSCAN, patient lookup, summary	CSV, slider values, patient values	Interactive charts and tables

6.3 Important Code Snippets

Only the main parts of the code are shown here. The complete code is in Appendix B and is submitted separately.

Cleaning and scaling:

```
MISSING_COLS = ["Glucose","BloodPressure","SkinThickness","Insulin","BMI"]
df[MISSING_COLS] = df[MISSING_COLS].replace(0, np.nan)
X_imp = pd.DataFrame(SimpleImputer(strategy="median").fit_transform(df[FEATS]),
columns=FEATS)
X = StandardScaler().fit_transform(X_imp)
```

K-Means over k = 2 to 8, and the final model:

```
for k in range(2, 9):
    km = KMeans(n_clusters=k, n_init=10, random_state=42).fit(X)
    sil = silhouette_score(X, km.labels_)
    dbi = davies_bouldin_score(X, km.labels_)
km = KMeans(n_clusters=3, n_init=10, random_state=42).fit(X)
```

DBSCAN grid search and the final model:

```
for eps in np.arange(1.0, 3.01, 0.1):
    for ms in (5, 8, 12, 16):
        labels = DBSCAN(eps=eps, min_samples=ms).fit(X).labels_
db = DBSCAN(eps=2.0, min_samples=16).fit(X)
```

PCA for the two-dimensional plots:

```
pca = PCA(n_components=2, random_state=42).fit(X)
Z = pca.transform(X)
```

7. RESULTS AND DISCUSSION

7.1 Exploratory Data Analysis

Before clustering, the data was examined. Insulin is strongly skewed to the right (skewness 3.38 after cleaning), as are the diabetes pedigree function (1.92) and age (1.13). Pregnancies and age are correlated (0.54), skin thickness and BMI are correlated (0.54), and glucose and insulin are correlated (0.42). Because several features move together, the data is expected to form overlapping groups and not sharply separated ones.

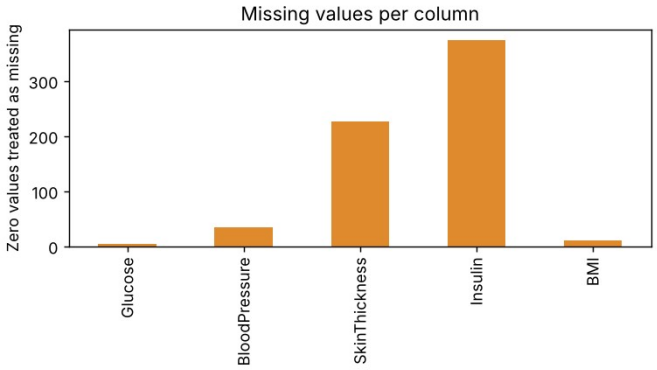


Figure 7.1: Zero values treated as missing, by column

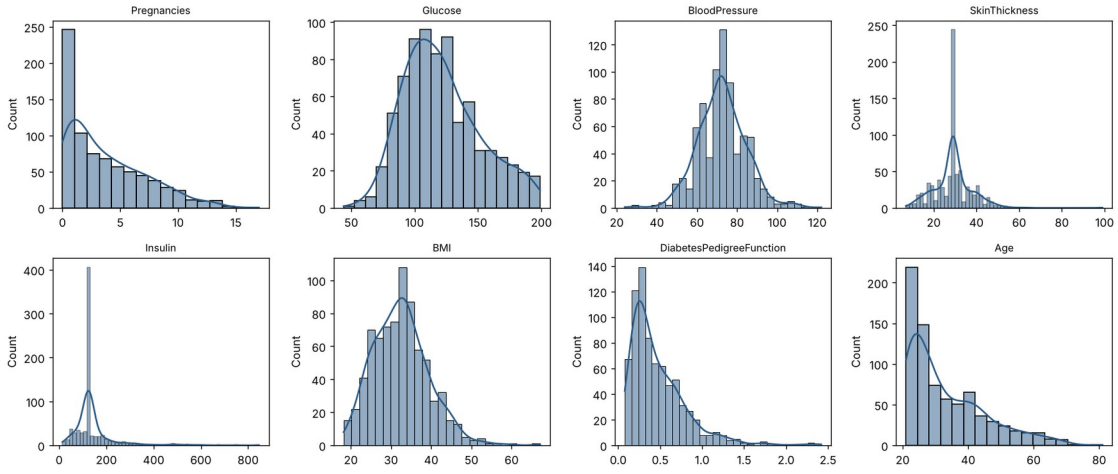


Figure 7.2: Distribution of each feature after cleaning

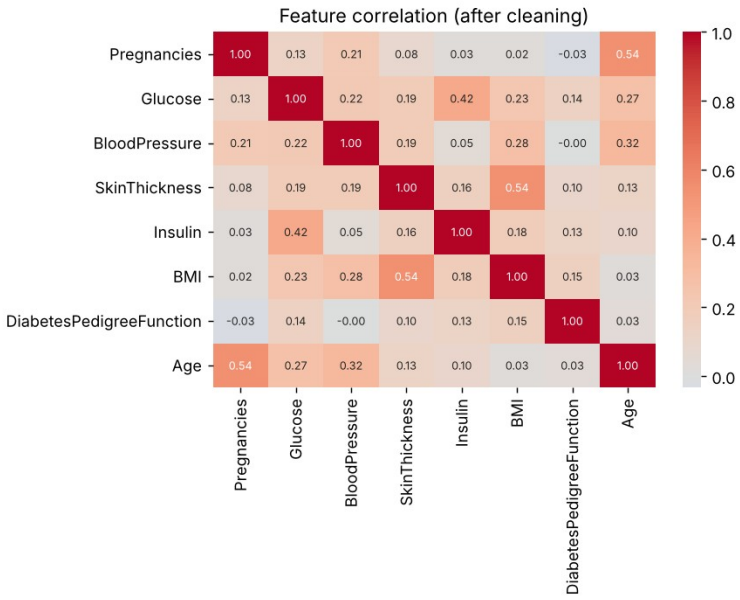


Figure 7.3: Correlation between the features

7.2 Model Performance

Model	Clusters / noise	Silhouette	Davies-Bouldin	Stability (ARI, 10 seeds)	Remarks
K-Means, k = 2	2	0.196	1.920	0.997	Highest silhouette; splits younger lower-risk from older higher-risk patients
K-Means, k = 3 (chosen)	3	0.195	1.834	0.991	Three balanced groups that can be described clearly
K-Means, k = 4	4	0.190	1.617	0.959	Lower Davies-Bouldin but less stable
DBSCAN, eps 2.0, min_samples 16	1 cluster + 88 noise	Not defined (one cluster)	Not defined	Not applicable	One dense group and outliers

Table 7.2 compares the methods. The full K-Means results for k = 2 to 8 are in Appendix C. The silhouette scores are almost the same for k = 2 and k = 3 and then fall, and there is no sharp bend in the inertia curve, so k = 3 was chosen as a judgement for interpretability and balanced group sizes, not because the metrics clearly prefer it.



Figure 7.4: Elbow, silhouette and Davies-Bouldin for k = 2 to 8

Average profile of each K-Means group (original units):

Cluster	Patients	Age	Pregnancies	Glucose	BMI	Insulin	Diabetes rate
0	248	46	7.4	131	32.8	140	52%
1	335	26	2.3	107	28.5	113	13%
2	185	29	1.9	137	39.1	192	52%

Cluster 0 (248 patients): older (average 46), many pregnancies (7.4), higher blood pressure; diabetes rate 52%. Cluster 1 (335 patients): younger (average 26), lowest glucose and BMI; diabetes rate 13%. Cluster 2 (185 patients): younger (average 29), highest BMI (39.2), skin thickness and insulin, high glucose; diabetes rate 52%.

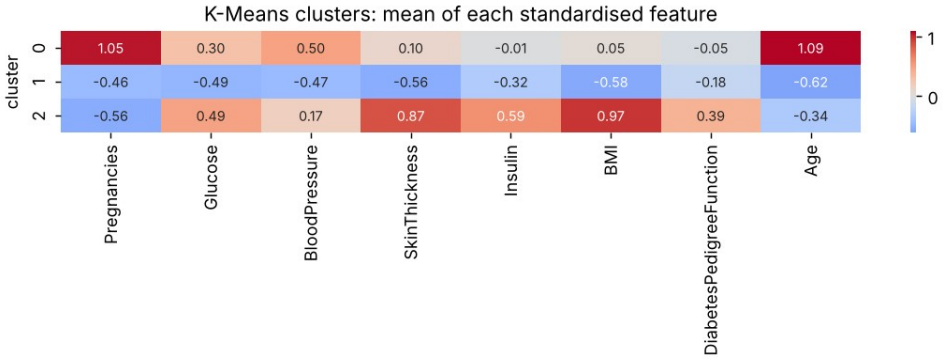


Figure 7.5: Average of each standardised feature in each K-Means cluster

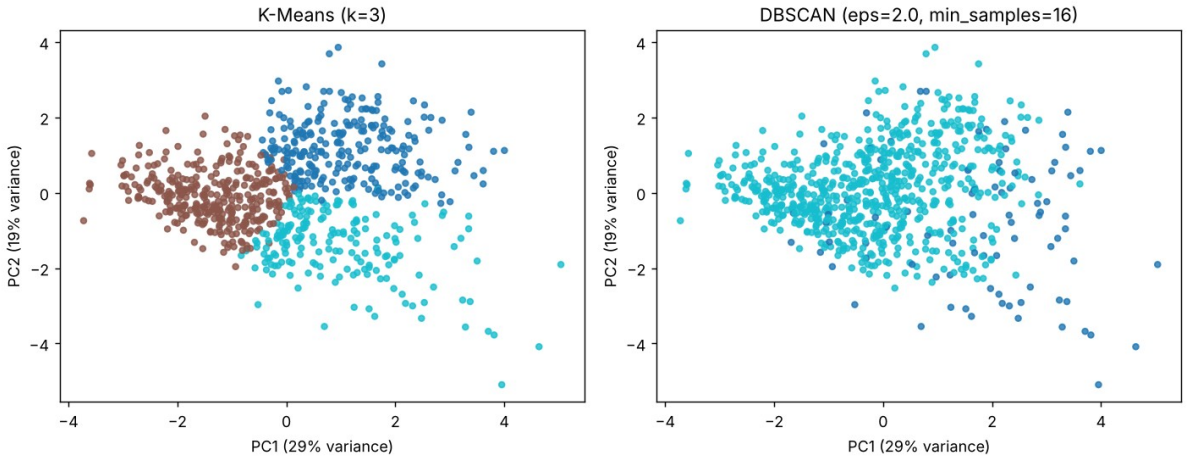


Figure 7.6: K-Means (left) and DBSCAN (right) shown on the first two principal components (47% of the variance)

7.3 Confusion Matrix / Error Analysis

Confusion matrices are not used because clustering has no true labels. Instead, the groups were checked against the diabetes label after clustering, and the main errors and weaknesses are described below.

Group	No diabetes	Diabetes	Total	Diabetes rate
K-Means cluster 0	120	128	248	52%
K-Means cluster 1	291	44	335	13%
K-Means cluster 2	89	96	185	52%
DBSCAN noise (-1)	40	48	88	55%
DBSCAN cluster 0	460	220	680	32%

Table 7.3: Cross-tabulation of clusters and the diabetes label (the label was not used to build the clusters).

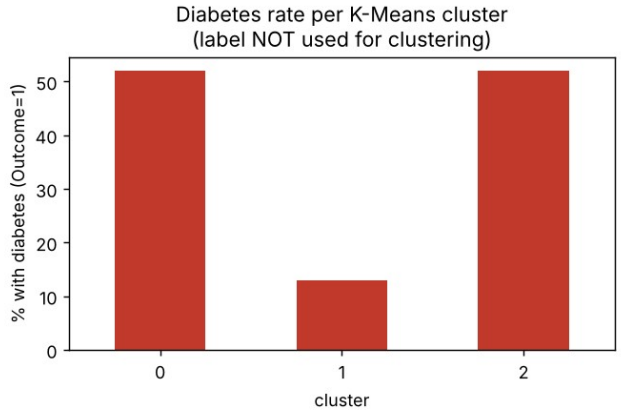


Figure 7.7: Diabetes rate in each K-Means cluster

- The clusters overlap. The silhouette score of 0.195 is low, and in Figure 7.6 the three groups touch each other.
- Insulin (374 values, 48.7%) and skin thickness (227 values, 29.6%) are heavily filled with the median, which may hide real differences between patients.
- Choosing k = 3 is a judgement, because k = 2 has almost the same silhouette score.
- Group membership says little about any single patient: 44 patients in the lowest-risk group (cluster 1) have diabetes, and 120 patients in cluster 0 do not.

7.4 Discussion

K-Means gave more useful segments than DBSCAN. The three groups differ in age, body measurements and diabetes rate (13% against 52%), and they are very stable (mean ARI 0.991, minimum 0.981 over ten random starts). The diabetes label was not used to build the groups, so the difference in diabetes rate is independent evidence that the groups carry clinical meaning.

The groups are not sharply separated. The data behaves more like one continuous cloud of patients that K-Means cuts into three regions than like three distinct types of patient.

DBSCAN found one dense group of 680 patients and 88 noise patients. The noise group has higher glucose (146), much higher insulin (266) and a higher pedigree function (0.81) than the main group, and a diabetes rate of 55% against 32%. DBSCAN is therefore better used here to flag unusual high-risk patients. Only two settings in the whole grid produced two clusters, and in both the second cluster had 4 or 5 patients, which is a pocket of outliers and not a patient segment.

The results describe patients with similar measurements. They are not diagnoses and must not be used for medical decisions.

7.5 Web Application Output

A Streamlit web application was built so that the results can be explored interactively. Its pages are shown below.

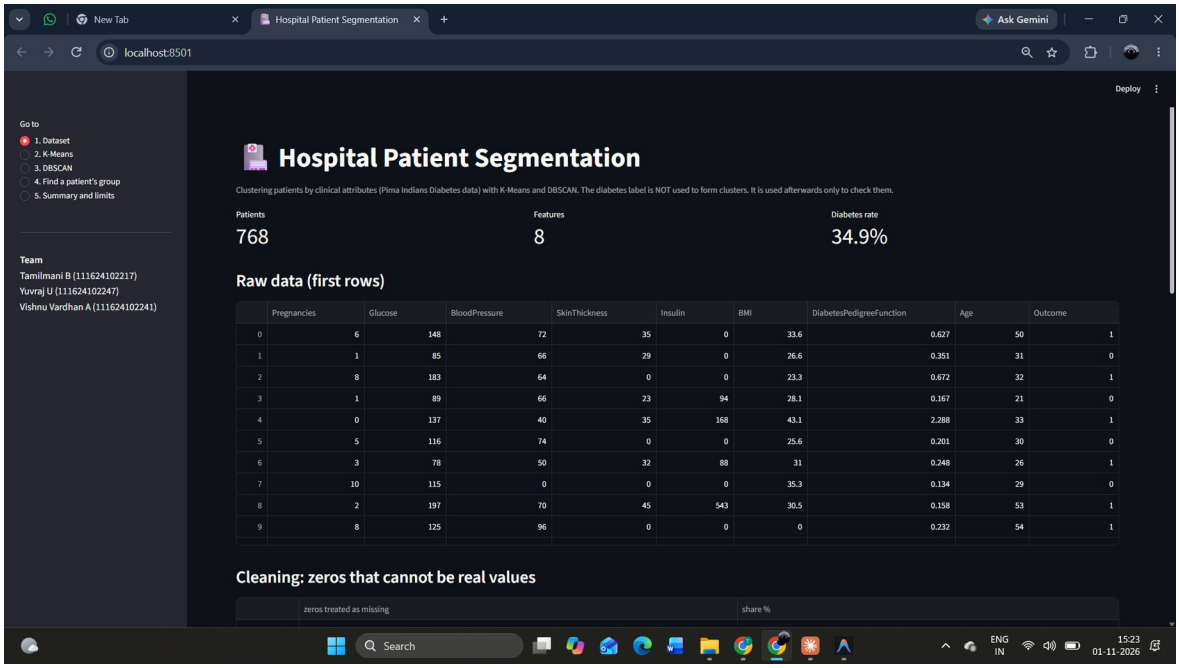


Figure 7.8: Dataset page (patients, features, diabetes rate and the first rows of data)

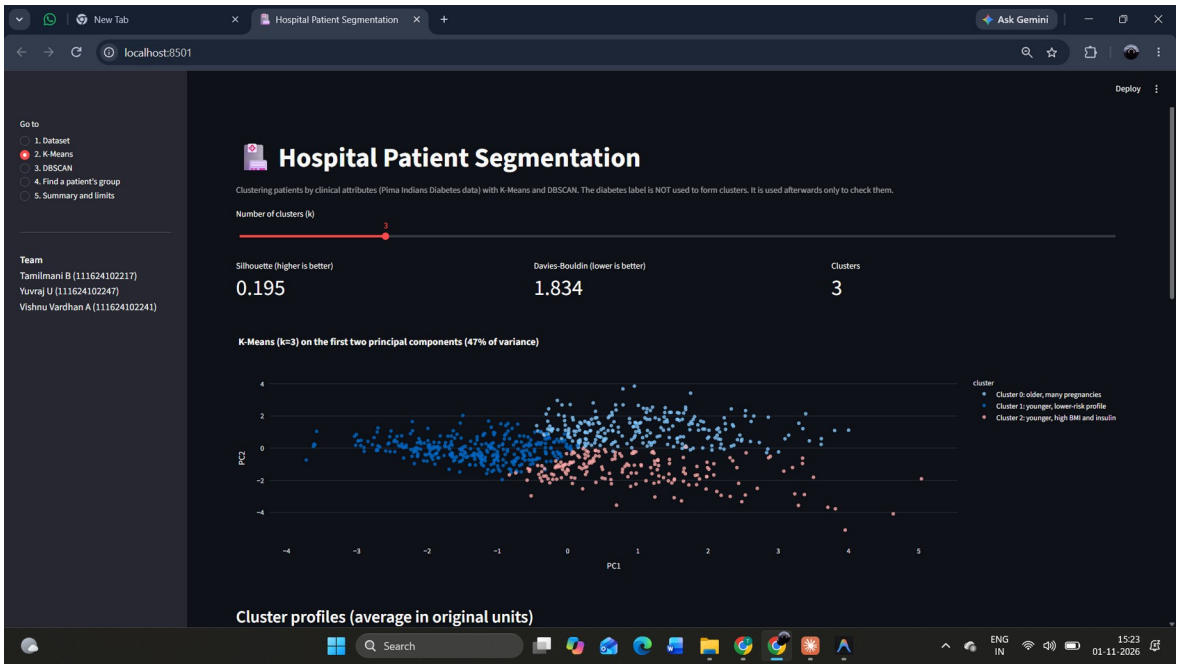


Figure 7.9: K-Means page with the k slider, quality scores and PCA scatter plot

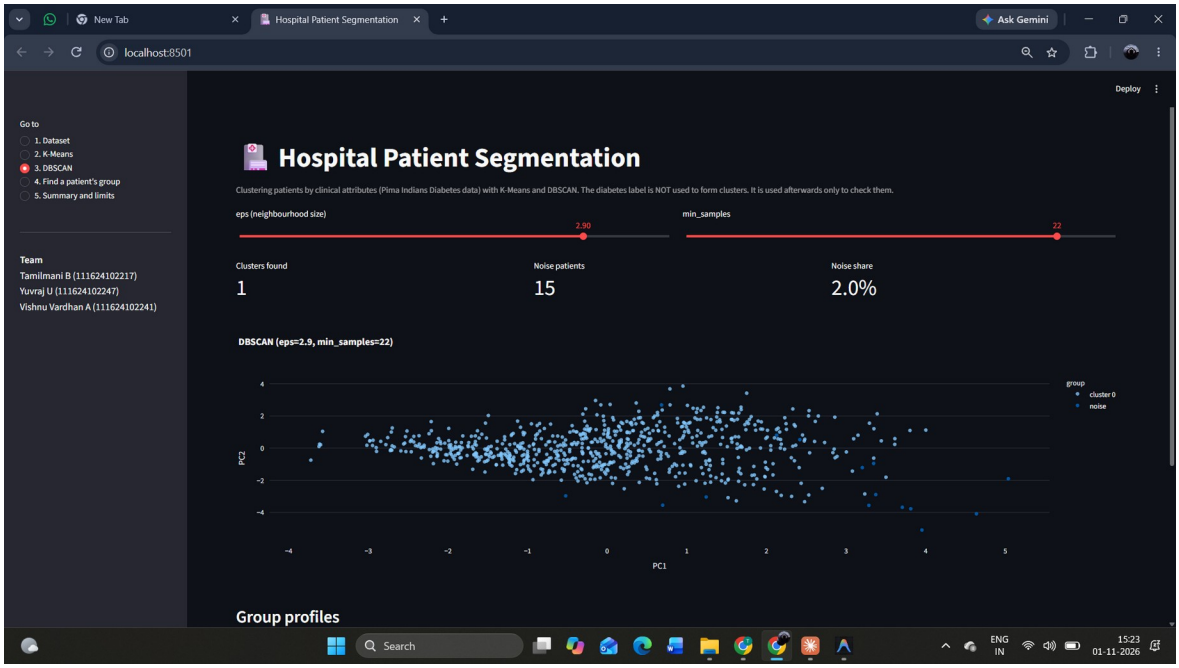


Figure 7.10: DBSCAN page with eps and min_samples sliders (shown here at eps = 2.9 and min_samples = 22)

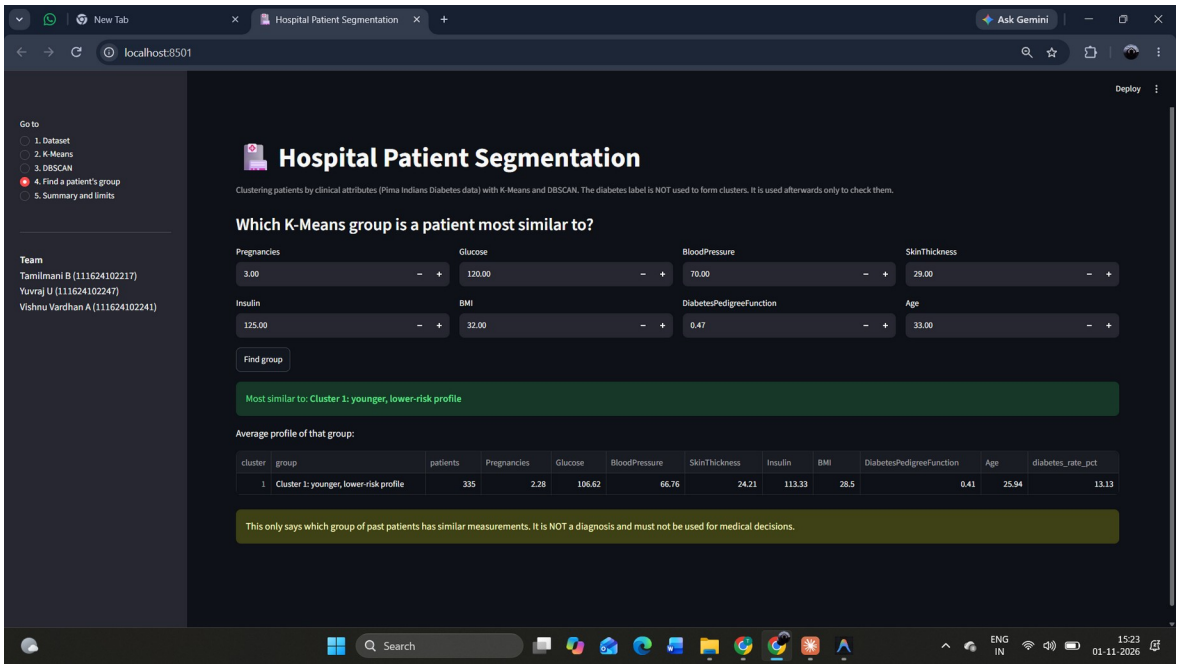


Figure 7.11: Patient lookup page, which places a new patient in the most similar K-Means group

8. CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

Using K-Means with k = 3, the 768 patients fell into three stable and interpretable groups with very different diabetes rates (52%, 13% and 52%), although the groups overlap (silhouette score 0.195). DBSCAN found one main group and 88 unusual patients and did not give separate segments. All four objectives were met: the data was cleaned and scaled, both algorithms were applied, they were evaluated with several measures and compared, and the results were checked against the diabetes label and presented in a web application. For patient segmentation on this data K-Means is the better method, and DBSCAN is useful for spotting outliers.

8.2 Limitations

The data covers only women aged 21 or older of Pima Indian heritage, so the groups may not hold for other patients. There are only 768 patients. Insulin and skin thickness are heavily filled with the median. Clusters describe similarity of measurements and are not diagnoses. The choice of k = 3 is a judgement. No doctor checked whether the groups are clinically useful.

8.3 Future Scope

- Try Gaussian mixture models and hierarchical clustering, which can handle overlapping groups.
- Handle missing insulin and skin thickness with better methods (for example model-based imputation) and compare the clusters.
- Test the approach on a larger and more varied hospital dataset, and ask clinicians to judge whether the groups are useful.

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10. APPENDICES

Appendix A: Dataset Link and Data Sample

Dataset URL: https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database
First five records of the data used:

Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
6.0	148.0	72.0	35.0	0.0	33.6	0.627	50.0	1.0
1.0	85.0	66.0	29.0	0.0	26.6	0.351	31.0	0.0
8.0	183.0	64.0	0.0	0.0	23.3	0.672	32.0	1.0
1.0	89.0	66.0	23.0	94.0	28.1	0.167	21.0	0.0
0.0	137.0	40.0	35.0	168.0	43.1	2.288	33.0	1.0

Appendix B: Source Code Repository

Repository URL: https://github.com/TharunS2006/CleanEnergy/tree/claude/new-session-uqjqaj/ml_mini_project

Appendix C: Additional Results

K-Means results for k = 2 to 8:

k	Inertia	Silhouette	Davies-Bouldin	Calinski-Harabasz
2	4975.0	0.196	1.920	180.0
3	4335.7	0.195	1.834	159.5
4	3912.7	0.190	1.617	145.2
5	3678.7	0.177	1.609	127.8
6	3474.5	0.168	1.732	117.1
7	3307.3	0.123	1.847	108.8
8	3145.5	0.146	1.648	103.5

DBSCAN grid (selected rows). Smallest cluster is the size of the smallest cluster as a share of all patients:

eps	min_samples	Clusters	Noise	Smallest cluster
1.0	5	6	72.5%	0.5%
1.5	8	1	28.1%	71.9%
2.0	5	2	6.0%	0.7%
2.0	16	1	11.5%	88.5%
2.1	5	2	5.2%	0.5%
2.5	16	1	3.9%	96.1%

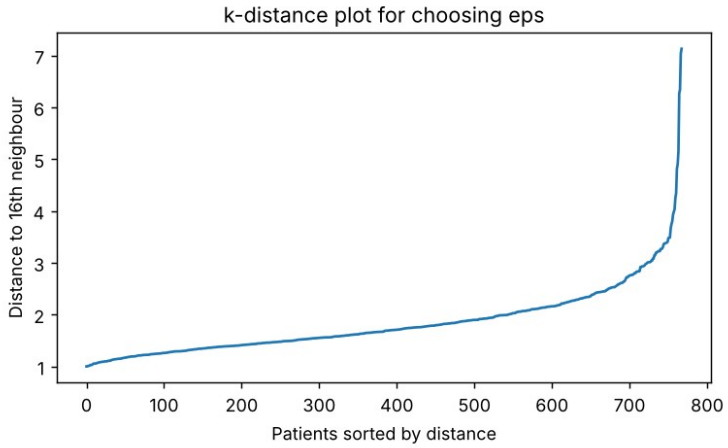


Figure C.1: k-distance plot (distance to the 16th nearest neighbour)

Appendix D: Individual Contribution

Student Name / Register No.	Task Assigned	Task Completed	Signature
Tamil Mani B / 111624102217	Data cleaning, exploratory analysis and K-Means	Zero-value handling, imputation, scaling, EDA plots, choice of k	
A Vishnu Vardhan / 111624102241	DBSCAN, evaluation and comparison	k-distance plot, grid search, metrics, stability test	
Yuvaraj U / 111624102247	Web application, visualisation and report	Streamlit app, PCA plots, report writing	

FACULTY EVALUATION RUBRIC

Assessment Component	Marks	Evaluation Criteria	Awarded Marks
Problem Definition and Objectives	10	Clarity, relevance, measurable objectives	
Literature Survey and Dataset Understanding	10	Relevant sources, dataset suitability, documentation	
Preprocessing and EDA	15	Data cleaning, visualization, feature preparation	
Algorithm Implementation	20	Correct implementation and technical understanding	
Evaluation and Results	20	Appropriate metrics, interpretation, comparison	
Report Quality	10	Organization, formatting, references, originality	
Presentation and Viva	10	Communication, demonstration, responses	
Team Contribution / Individual Work	5	Evidence of individual contribution	
Total	100		

STUDENT SUBMISSION CHECKLIST

☐ Approved project title and guide details included

☐ Abstract and keywords included

☐ Problem statement and objectives clearly stated

☐ Dataset source and Kaggle link included

☐ Dataset characteristics and data dictionary provided

☐ Preprocessing steps documented

☐ At least one machine learning algorithm implemented

☐ Evaluation metrics justified and reported

☐ Results supported with tables/plots

☐ Limitations and future scope discussed

☐ References and dataset attribution included

☐ Source code/notebook submitted

☐ Plagiarism and ethical considerations addressed

☐ Final report proofread and formatted